

Advanced Microeconomics

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Problem Set 1: Math for Economists, Preferences SOLUTION

1 Important rules

$$\begin{array}{ll} f(x) = x^n & \rightarrow f'(x) = n \cdot x^{n-1} \\ f(x) = c \cdot y(x) & \rightarrow f'(x) = c \cdot y'(x) \\ f(x) = g(x) + h(x) & \rightarrow f'(x) = g'(x) + h'(x) \\ f(x) = g(x) - h(x) & \rightarrow f'(x) = g'(x) - h'(x) \\ f(x) = g(x) \cdot h(x) & \rightarrow f'(x) = g'(x) \cdot h(x) + g(x) \cdot h'(x) \\ f(x) = \frac{g(x)}{h(x)} & \rightarrow f'(x) = \frac{h(x) \cdot g'(x) - g(x) \cdot h'(x)}{(h(x))^2} \\ f(x) = g(h(x)) & \rightarrow f'(x) = g'(h(x)) \cdot h'(x) \end{array}$$

1. Math

a) Determine the first derivative of the following functions (according to the independent variable given in each case):

$$\begin{array}{ll} a) f(x) = (x+2)\sqrt{x} & \rightarrow f'(x) = \frac{3x+2}{2\sqrt{x}} \\ b) f(x) = \frac{x^2+2}{2x} & \rightarrow f'(x) = \frac{x^2-2}{2x^2} \\ c) f(x) = \sqrt{3x^2-1} & \rightarrow f'(x) = \frac{3x}{\sqrt{3x^2-1}} \\ d) k(x) = x^a \cdot e^{-ax} & \rightarrow k'(x) = ae^{-ax} \cdot (x^{a-1} - x^a) \\ e) P(w) = \left(\ln \frac{w^2+1}{e^w} \right)^{20} & \rightarrow P'(w) = 20 \left(\ln \frac{w^2+1}{e^w} \right)^{19} \cdot \left(\frac{2w}{w^2+1} - 1 \right) \\ f) Q(s) = \ln \sqrt{\frac{1+s^4}{6+s^2}} & \rightarrow Q'(s) = \frac{2s^3}{1+s^4} - \frac{s}{6+s^2} \end{array}$$

b) Find the first through third derivatives of the following functions:

$$\begin{aligned}
 a) \quad & f(x) = x^{10} \rightarrow f'(x) = 10x^9; \quad f''(x) = 90x^8; \quad f'''(x) = 720x^7 \\
 b) \quad & g(y) = y \cdot \ln y \rightarrow g'(y) = \ln y + 1; \quad g''(y) = \frac{1}{y}; \quad g'''(y) = -\frac{1}{y^2} \\
 c) \quad & p(t) = t \cdot e^t \rightarrow p'(t) = e^t(t+1); \quad p''(t) = e^t(t+2); \quad p'''(t) = e^t(t+3) \\
 d) \quad & h(z) = \frac{z+1}{(z-1)^2} \rightarrow h'(z) = -\frac{z+3}{(z-1)^3}; \quad h''(z) = 2\frac{z+5}{(z-1)^4}; \quad h'''(z) = -6\frac{z+7}{(z-1)^5} \\
 e) \quad & k(r) = e^{\frac{1}{r}} \rightarrow k'(r) = -\frac{1}{r^2}e^{\frac{1}{r}}; \quad k''(r) = \left(\frac{2}{r^3} + \frac{1}{r^4}\right)e^{\frac{1}{r}}; \quad k'''(r) = \left(-\frac{6}{r^4} - \frac{6}{r^5} - \frac{1}{r^6}\right)e^{\frac{1}{r}}
 \end{aligned}$$

c) Form all the first-order partial derivatives of the following functions.

$$\begin{aligned}
 a) \quad & f(x, y) = 3x^2 - 4y^2 + 5xy + 4y \rightarrow f'_x(x, y) = 6x + 5y; \quad f'_y(x, y) = -8y + 5x + 4 \\
 b) \quad & f(x, y) = \frac{x^4 - 3x^2y}{3x + 2y^2} \quad \left(x \neq -\frac{2}{3}y^2\right) \rightarrow f'_x(x, y) = \frac{9x^4 - 9x^2y + 8x^3y^2 - 12xy^3}{(3x + 2y^2)^2}; \\
 & f'_y(x, y) = \frac{-9x^3 + 6x^2y^2 - 4x^4y}{(3x + 2y^2)^2} \\
 c) \quad & f(x, y) = 2y^{3x} \cdot \ln \frac{x}{y}; \quad (x, y > 0) \rightarrow f'_x(x, y) = 2y^{3x} \cdot \left(3 \ln y \ln \frac{x}{y} + \frac{1}{x}\right); \\
 & f'_y(x, y) = 2y^{3x-1} \cdot \left(3x \ln \frac{x}{y} - 1\right)
 \end{aligned}$$

d) Find all partial derivatives of first and second order for the following functions:

$$\begin{aligned}
 a) \quad & K(x, y) = 150 + 3x^2 - 3xy - 6x + \frac{3}{2}y^3 + 3y \rightarrow K'_x(x, y) = 6x - 3y - 6; \\
 & K'_y(x, y) = -3x + \frac{9}{2}y^2 + 3; \quad K''_{xx}(x, y) = 6; \quad K''_{yy}(x, y) = 9y; \quad K''_{xy} = -3 \\
 b) \quad & P(A, K) = \frac{1}{2}A^{\frac{1}{3}} \cdot K^{\frac{2}{3}} \rightarrow P'_A(A, K) = \frac{1}{6}A^{-\frac{2}{3}}K^{\frac{2}{3}} = \frac{1}{6}\left(\frac{K}{A}\right)^{\frac{2}{3}}; \quad P'_K(A, K) = \frac{1}{3}A^{\frac{1}{3}}K^{-\frac{1}{3}} = \frac{1}{3}\left(\frac{A}{K}\right)^{\frac{1}{3}}; \\
 & P''_{AA}(A, K) = -\frac{1}{9}A^{-\frac{5}{3}}K^{\frac{2}{3}}; \quad P''_{KK}(A, K) = -\frac{1}{9}A^{\frac{1}{3}}K^{-\frac{4}{3}}; \quad P''_{AK} = \frac{1}{9}A^{-\frac{2}{3}}K^{-\frac{1}{3}} \\
 c) \quad & K(x, y) = x + \ln \frac{y}{x} \quad (x, y > 1) \rightarrow K'_x(x, y) = 1 - \frac{1}{x}; \quad K'_y(x, y) = \frac{1}{y}; \\
 & K''_{xx}(x, y) = \frac{1}{x^2}; \quad K''_{yy}(x, y) = -\frac{1}{y^2}; \quad K''_{xy}(x, y) = 0 \\
 d) \quad & K(x, y) = \frac{1}{2}(x-a)^2 \cdot (y-1) + \frac{b}{2}y^2 - 3y + 20 \quad (a, b \in \mathbb{R}) \rightarrow K'_x(x, y) = (x-a)(y-1); \\
 & K'_y(x, y) = \frac{1}{2}(x-a)^2 + by - 3; \quad K''_{xx}(x, y) = y-1; \quad K''_{yy}(x, y) = b; \quad K''_{xy}(x, y) = x-a
 \end{aligned}$$

2. Preferences

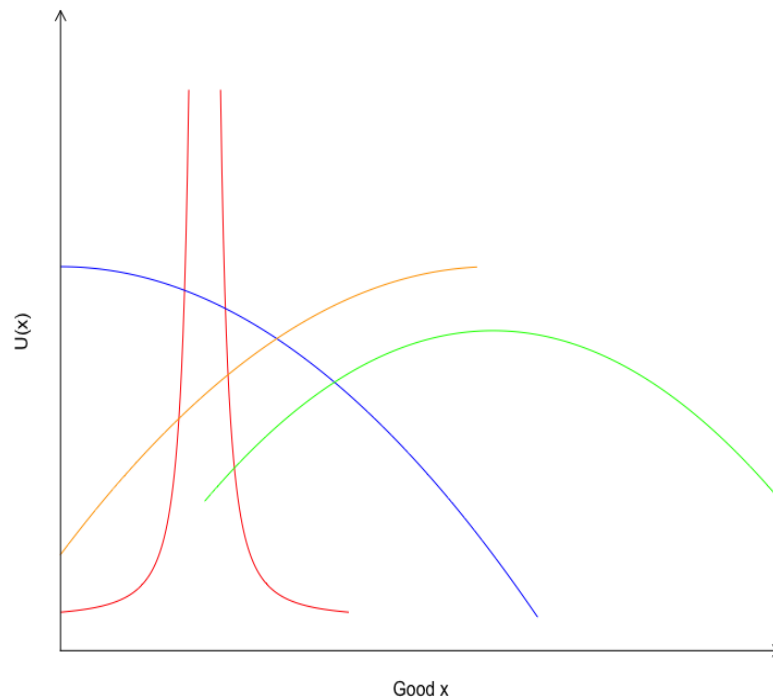
a) Provide brief definitions for the following preference relations:

$\succsim$ $\succ$ $\sim$
 weakly preferred strictly preferred indifferent

b) Consider a rational preference relation ($\succsim$). Describe verbally which requirements (axioms) $\succsim$ has to satisfy.

Solution: The necessary requirements for a rational preference relation are transitivity and completeness

c) The Figure below shows curves of utility dependent on a good x. Assess whether the utility functions represent rational preferences. Explain which preferences satisfy the axioms of completeness, transitivity and monotonicity.



Solution:

- Red: Preferences are not complete. Preferences are not defined for the quantity of x between the two red lines. Red also violates the axioms of monotonicity and transitivity.
- Green: Monotonicity and transitivity are violated.
- Blue: Preferences are complete and transitive; $x_1 \succ x_2$ and $x_2 \succ x_3$, therefore also $x_1 \succ x_3$. Monotonicity is violated, more is not better $\rightarrow$ Bad good, less is better.
- Orange: No axioms are violated.

d) Sketch the indifference curves of the preferences depicted in the examples below. Name which type of goods are presented. Highlight the direction in which utility increases (e.g. $IC1 < IC2$).

- i) Anne loves cheese. In fact, she doesn't care about which kind of cheese she eats. Her closest supermarket offers only two kinds of cheese, Mozzarella and Parmegiano.
- ii) Mary loves milk. However, she only enjoys milk if she also has cookies to dip in the milk. In fact, Mary insists on eating three cookies for every glass of milk she drinks. What is her utility function?
- iii) For Christmas employees of Papa Joe's Pizza company receive coupons for books and visiting a stake house restaurant. However, John is vegetarian and therefore doesn't care about visits to stake restaurants. He does enjoy reading books though.
- iv) More and more coal factories settle around the city of H. On one hand, this does benefit the population as the factories bring a lot of jobs. However, the increased industrial activity also causes air pollution.
- v) Tom is an introverted person. Mostly he enjoys places where there are no cars and no people. Are the indifference curves convex or concave and why?

Solution:

- i) Anne is indifferent between the two. $x \sim y$. The goods are perfect substitutes.
- ii) The goods are perfect complements. The utility function would be: $U(x, y) = \min(x, \frac{y}{3})$, where x refers to glasses of milk and y refers to the quantity of cookies.
- iii) Here, book coupons are normal goods, whereas stake house coupons are considered bad goods in this context.
- iv) Additional jobs are normal goods whereas air pollution is a bad good.
- v) Tom dislikes cars and people. Both are bad goods.